# Amazon SageMaker Resources in AWS Marketplace

Amazon SageMaker integrates with AWS Marketplace, enabling developers to charge other Amazon SageMaker users for the use of their algorithms and model packages. AWS Marketplace is a curated digital catalog that makes it easy for customers to find, buy, deploy, and manage third-party software and services that customers need to build solutions and run their businesses. AWS Marketplace includes thousands of software listings in popular categories, such as security, networking, storage, machine learning, business intelligence, database, and DevOps. It simplifies software licensing and procurement with flexible pricing options and multiple deployment methods. For information, see AWS Marketplace Documentation.

## Topics

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## Amazon SageMaker Algorithms

An algorithm enables you to perform end-to-end machine learning. It has two logical components: training and inference. Buyers can use the training component to create training jobs in Amazon SageMaker and build a machine learning model. Amazon SageMaker saves the model artifacts generated by the algorithm during training to an Amazon S3 bucket. For more information, see Train a Model with Amazon SageMaker (p. 4).

Buyers use the inference component with the model artifacts generated during a training job to create a deployable model in their Amazon SageMaker account. They can use the deployable model for real-time inference by using Amazon SageMaker hosting services. Or, they can get inferences for an entire dataset by running batch transform jobs. For more information, see Deploy a Model in Amazon SageMaker (p. 7).

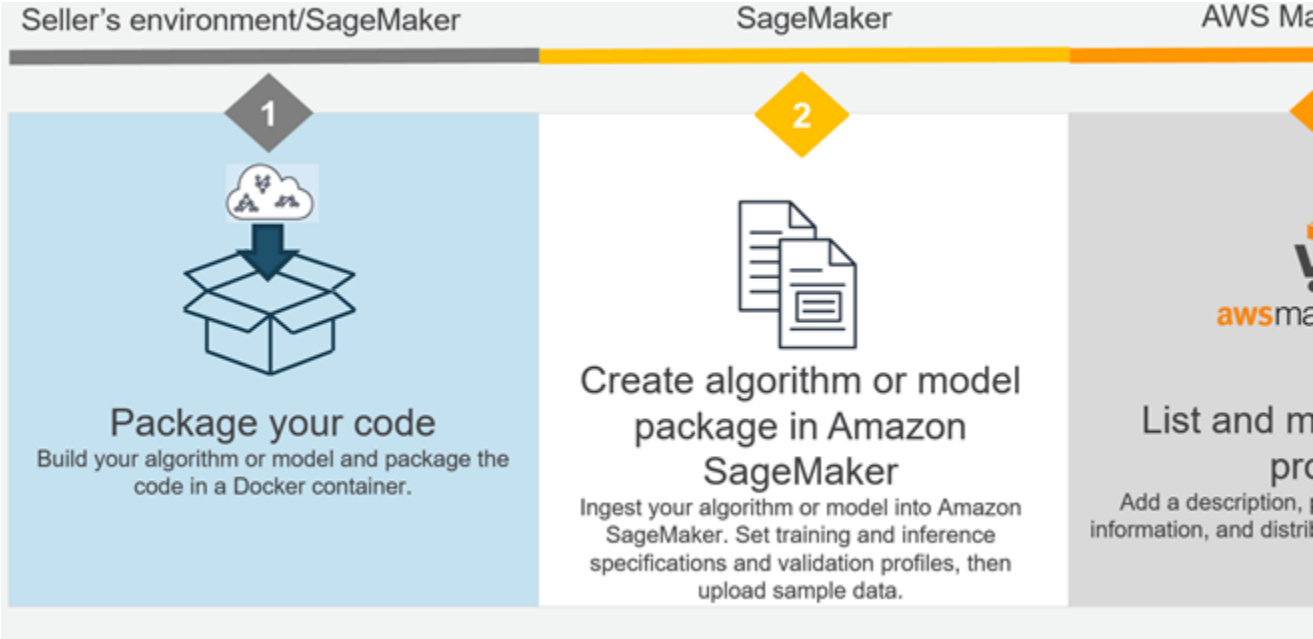
## Amazon SageMaker Model Packages

Buyers use a model package to build a deployable model in Amazon SageMaker. They can use the deployable model for real-time inference by using Amazon SageMaker hosting services. Or, they can get inferences for an entire dataset by running batch transform jobs. For more information, see Deploy a Model in Amazon SageMaker (p. 7). As a seller, you can build your model artifacts by training in Amazon SageMaker, or you can use your own model artifacts from a model that you trained outside of Amazon SageMaker. You can charge buyers for inference.

# Sell Amazon SageMaker Algorithms and Model Packages

Selling Amazon SageMaker algorithms and model packages is a three-step process:

1. Develop your algorithm or model, and package it in a Docker container. For information, see Develop Algorithms and Models in Amazon SageMaker (p. 429).
2. Create an algorithm or model package resource in Amazon SageMaker. For information, see Create Algorithm and Model Package Resources (p. 413).
3. Register as a seller on AWS Marketplace and list your algorithm or model package on AWS Marketplace. For information about registering as a seller, see Getting Started as a Seller in the *User Guide for AWS Marketplace Providers*. For information about listing and monetizing your algorithms and model packages, see Listing Algorithms and Model Packages in AWS Marketplace for Machine Learning in the *User Guide for AWS Marketplace Providers*.




## Topics

- Develop Algorithms and Models in Amazon SageMaker (p. 429)
- Create Algorithm and Model Package Resources (p. 413)
- List Your Algorithm or Model Package on AWS Marketplace (p. 431)

## Develop Algorithms and Models in Amazon SageMaker

Before you can create algorithm and model package resources to use in Amazon SageMaker or list on AWS Marketplace, you have to develop them and package them in Docker containers.

> **Note**
> When algorithms and model packages are created for listing on AWS Marketplace, Amazon SageMaker scans the containers for security vulnerabilities on supported operating systems. Only the following operating system versions are supported:
>
> - Debian: 6.0, 7, 8, 9, 10
> - Ubuntu: 12.04, 12.10, 13.04, 14.04, 14.10, 15.04, 15.10, 16.04, 16.10, 17.04, 17.10, 18.04, 18.10
> - CentOS: 5, 6, 7
> - Oracle Linux: 5, 6, 7
> - Alpine: 3.3, 3.4, 3.5
> - Amazon Linux

**Topics**

- Develop Algorithms in Amazon SageMaker (p. 430)
- Develop Models in Amazon SageMaker (p. 430)

# Develop Algorithms in Amazon SageMaker

An algorithm should be packaged as a docker container and stored in Amazon ECR to use it in Amazon SageMaker. The Docker container contains the training code used to run training jobs and, optionally, the inference code used to get inferences from models trained by using the algorithm.

For information about developing algorithms in Amazon SageMaker and packaging them as containers, see Use Your Own Algorithms or Models with Amazon SageMaker (p. 384). For a complete example of how to create an algorithm container, see the sample notebook at https://github.com/awslabs/amazon-sagemaker-examples/blob/master/advanced_functionality/scikit_bring_your_own/scikit_bring_your_own.ipynb. You can also find the sample notebook in an Amazon SageMaker notebook instance. The notebook is in the **Advanced Functionality** section, and is named `scikit_bring_your_own.ipynb`. For information about using the sample notebooks in a notebook instance, see Use Example Notebooks (p. 42).

Always thoroughly test your algorithms before you create algorithm resources to publish on AWS Marketplace.

> **Note**
> When a buyer subscribes to your containerized product, the Docker containers run in an isolated (internet-free) environment. When you create your containers, do not rely on making outgoing calls over the internet. Calls to AWS services are also not allowed.

# Develop Models in Amazon SageMaker

A deployable model in Amazon SageMaker consists of inference code, model artifacts, an IAM role that is used to access resources, and other information required to deploy the model in Amazon SageMaker. Model artifacts are the results of training a model by using a machine learning algorithm. The inference code must be packaged in a Docker container and stored in Amazon ECR. You can either package the model artifacts in the same container as the inference code, or store them in Amazon S3.

You create a model by running a training job in Amazon SageMaker, or by training a machine learning algorithm outside of Amazon SageMaker. If you run a training job in Amazon SageMaker, the resulting model artifacts are available in the `ModelArtifacts` field in the response to a call to the DescribeTrainingJob (p. 744) operation. For information about how to develop an Amazon SageMaker model container, see Use Your Own Inference Code (p. 408). For a complete example of how to create

a model container from a model trained outside of Amazon SageMaker, see the sample notebook at https://github.com/awslabs/amazon-sagemaker-examples/blob/master/advanced_functionality/xgboost_bring_your_own_model/xgboost_bring_your_own_model.ipynb. You can also find the sample notebook in an Amazon SageMaker notebook instance. The notebook is in the **Advanced Functionality** section, and is named `xgboost_bring_your_own_model.ipynb`. For information about using the sample notebooks in a notebook instance, see Use Example Notebooks (p. 42).

Always thoroughly test your models before you create model packages to publish on AWS Marketplace.

> **Note**
> When a buyer subscribes to your containerized product, the Docker containers run in an isolated (internet-free) environment. When you create your containers, do not rely on making outgoing calls over the internet. Calls to AWS services are also not allowed.

## List Your Algorithm or Model Package on AWS Marketplace

After creating and validating your algorithm or model in Amazon SageMaker, list your product on AWS Marketplace. The listing process makes your products available in the AWS Marketplace and the Amazon SageMaker console.

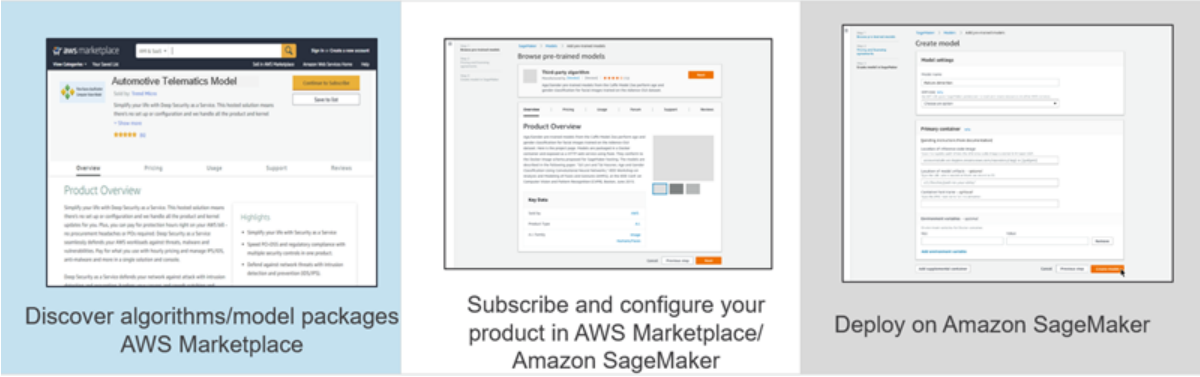
To list products on AWS Marketplace, you must be a registered seller. To register, use the self-registration process from the AWS Marketplace Management Portal (AMMP). For information, see Getting Started as a Seller in the *User Guide for AWS Marketplace Providers*. When you start the product listing process from the Amazon SageMaker console, we check your seller registration status. If you have not registered, we direct you to do so.

To start the listing process, do one of the following:

- From the Amazon SageMaker console, choose the product, choose **Actions**, and choose **Publish new ML Marketplace listing**. This carries over your product reference, the Amazon Resource Name (ARN), and directs you to the AMMP to create the listing.
- Go to ML listing process, manually enter the Amazon Resource Name (ARN), and start your product listing. This process carries over the product metadata that you entered when creating the product in Amazon SageMaker. For an algorithm listing, the information includes the supported instance types and hyperparameters. In addition, you can enter a product description, promotional information, and support information as you would with other AWS Marketplace products.

# Find and Subscribe to Algorithms and Model Packages on AWS Marketplace

With AWS Marketplace, you can browse and search for hundreds of machine learning algorithms and models in a broad range of categories, such as computer vision, natural language processing, speech recognition, text, data, voice, image, video analysis, fraud detection, predictive analysis, and more.




### To find algorithms on AWS Marketplace

1. Open the Amazon SageMaker console at https://console.aws.amazon.com/sagemaker/.
2. Choose **Algorithms**, then choose **Find algorithms**.

   This takes you to the AWS Marketplace algorithms page. For information about finding and subscribing to algorithms on AWS Marketplace, see Machine Learning Products in the *AWS Marketplace User Guide for AWS Consumers*.

### To find model packages on AWS Marketplace

1. Open the Amazon SageMaker console at https://console.aws.amazon.com/sagemaker/.
2. Choose **Model packages**, then choose **Find model packages**.

   This takes you to the AWS Marketplace model packages page. For information about finding and subscribing to model packages on AWS Marketplace, see Machine Learning Products in the *AWS Marketplace User Guide for AWS Consumers*.

# Use Algorithms and Model Packages

For information about using algorithms and model packages that you subscribe to in Amazon SageMaker, see Use Algorithm and Model Package Resources (p. 419).

> **Note**
> When you create a training job, inference endpoint, and batch transform job from an algorithm or model package that you subscribe to on AWS Marketplace, the training and inference containers do not have access to the internet. Because the containers do not have access to the internet, the seller of the algorithm or model package does not have access to your data.